

Zhenxiao Qi

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Education Background

- ✧ **University of California, Riverside (UCR)** 08/2018 - Present
 - ★ Degree expected: Ph.D.
 - ★ Major: Computer Science
- ✧ **University of California, Riverside (UCR)** 07/2017 - 06/2018
 - ★ Exchange student
 - ★ Major: Computer Science
- ✧ **Xidian University (XDU)** 07/2014 - 06/2017
 - ★ Degree: Bachelor of Engineering
 - ★ Major: Information Security

Working Experience

- ✧ **Software engineering intern @ Deepbits** 10/2021 - Present
 - ★ Host: Xunchao Hu
 - ★ Developing services to monitor and protect cloud VMs based on memory analysis
- ✧ **Summer research intern @ Google** 06/2021 - 09/2021
 - ★ Host: Mahesh Pisal
 - ★ Develop and deploy a logic inference-based profile generation approach for kernel memory analysis on Google cloud VMs
- ✧ **Graduate student researcher @ UCR** 08/2018 - Present
 - ★ Advisor: Prof. Heng Yin
- ✧ **Teaching assistant @ UCR** 09/2019 - 03/2021
 - ★ CS202 Advanced Operating System: Fall '19 / Winter '20
 - ★ CS153 Design of Operating systems: Fall '20
 - ★ CS100 Software Construction: Spring '20 / Winter '21

Research Experience

LogicMEM: A Logic Inference Approach to Automate Profile Generation for Binary-only Memory Forensics 04/2020 - 06/2021

Advisor: Prof. Heng Yin

- ★ The first binary-only approach that generates kernel profiles for downstream memory forensics.
- ★ Performed source code kernel evolution analysis and summarized kernel object invariants.
- ★ Widely tested on a range of Linux kernels from official distributions and showed high precision and recall.

SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets 07/2019 - 12/2020

Advisor: Prof. Heng Yin

- ★ A novel Spectre V1 gadget detection approach by enabling dynamic taint analysis on speculative execution paths.
- ★ The evaluation results show significantly improved precision and recall with reasonable efficiency.

- ★ Detected new Spectre V1 gadgets from real-world programs such as Brotli and Caffee.

QEMU Virtualization (KVM)/Emulation (TCG) Mode Switch

04/2018 - 06/2018

Advisor: Prof. Heng Yin

- ★ Achieved on-the-fly KVM/TCG mode switching on QEMU.
- ★ Enabled dynamic analysis that benefits from native speed of KVM when instrumentation is not needed.
- ★ Evaluated on common-used operating systems, such as Windows, Debian, Ubuntu, etc.

DECAF upgrade: porting to the latest version of QEMU

06/2018 - 09/2018

Advisor: Prof. Heng Yin

- ★ DECAF is a dynamic binary analysis platform based on QEMU.
- ★ Upgraded DECAF to the latest version of QEMU, and ported the functionalities of DECAF, such as Virtual Machine Introspection, taint analysis, etc.

Publications:

- ★ [NDSS'21] **Zhenxiao Qi**, Qian Feng, Yueqiang Cheng, Mengjia Yan, Peng Li, Heng Yin, and Tao Wei, SpecTaint: Speculative Taint Analysis for Discovering Spectre Gadgets, to appear in the Network and Distributed System Security Symposium, February 2021.
- ★ [RAID'19] Ali Davanian, **Zhenxiao Qi**, Yu Qu, and Heng Yin, DECAF++: Elastic Whole-System Dynamic Taint Analysis, in the 22nd International Symposium on Research in Attacks, Intrusions and Defenses, September 2019.

Professional Services:

Reviewer: Cybersecurity

Sub-reviewer: USENIX Security'21, DIMVA'19

Artifact Evaluation:

- ★ ACSAC'18 TIFF: Using Input Type Inference to Improve Fuzzing
- ★ ACSAC'19 Speculator: A Tool to Analyze Speculative Execution Attacks and Mitigations

Honors and Awards

- Outstanding Teaching Assistance in CSE Department 2019-2020
- Dean's Distinguished Fellowship, UC, Riverside 2018
- Outstanding Student Scholarship, Xidian University 2014-2017

Programming Skills:

Programming Language:

- Fluent in **C**, **Python**, **LaTeX**.
- Other language: **C++**, Java assembly language, MATLAB.

Operating Systems:

- **Linux**, Windows, XV6.

Tools and Platforms:

- AFL, Angr.
- QEMU, DECAF.
- TensorFlow, Caffee.
- Hadoop, Spark.